

ENGINEER & DESIGNER

Asher Straus

Manufacturing & Design Engineering · Northwestern University, Class of 2027

I design, analyze and build: from brake rotors machined on a mill, to a fixture for a million-unit-a-year line, to an AI system that plans sheet-metal bending. This is a selection of that work, with the full portfolio, more projects and photos online.

[View the full portfolio → strausasher.github.io/Portfolio](https://strausasher.github.io/Portfolio)

SELECTED PROJECTS

- | | | | |
|---|-------------------------------------|---|-----------------------------------|
| 1 | Formula SAE Brake System | 2 | Bionic Wrench Manufacturing |
| 3 | Water Guard | 4 | Performance Bicycle Brake Caliper |
| 5 | Smart Sheet Smith | 6 | ExtendIt |
| + | Hands-on shop work, and more online | | |

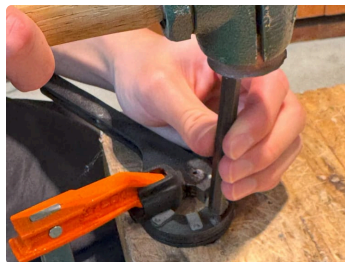
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Milling brake rotors



Torch-brazing copper



Hand-forming metal



Tesla repair

Formula SAE Brake System

Northwestern Formula Racing · 2024–Present

A brake system designed backwards from a target: stop the car at 1 G. I worked through the analytical force model, improved the team's MATLAB simulation, verified the rotors in ANSYS, then machined and finished them myself and routed the lines on the car.

1 G

deceleration target

~690 lbf

braking force at the tires

~42 lbf

driver pedal force

~16x

brake system gain

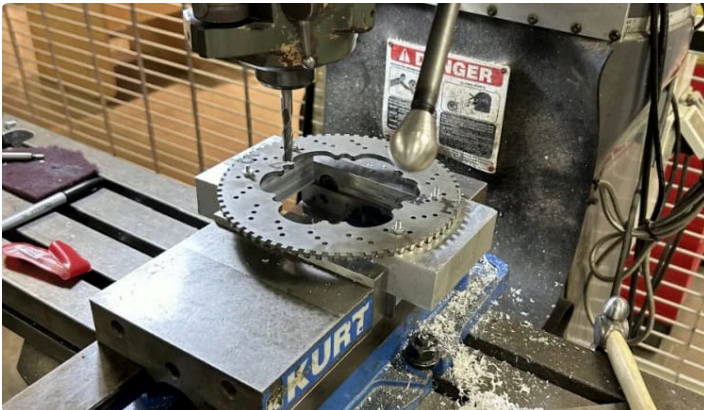
WHAT I DID

Sized the system for 1 G with an analytical pedal-to-tire force model (~42 lbf pedal to ~690 lbf at the tires).

Improved the team's MATLAB brake simulation so rotor diameter, master-cylinder bore and pedal ratio could be traded off quickly.

Checked rotor stress and heat in ANSYS (structural and thermal) before cutting any metal.

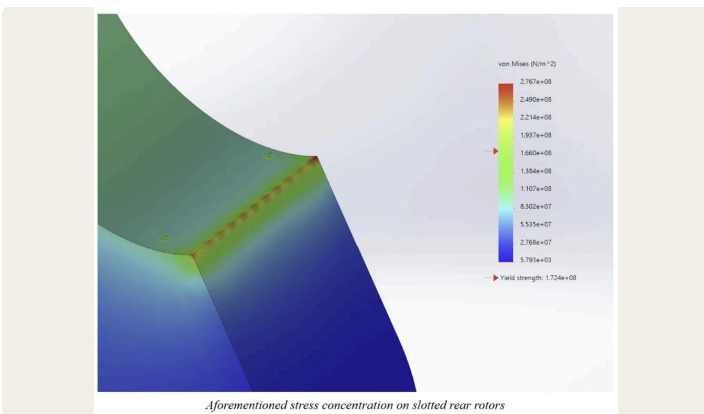
Machined and lathe-finished the rotors; routed lines and bled the system on the car.



Milling a rotor blank



Grinding the braking surface



Aforementioned stress concentration on slotted rear rotors

ANSYS rotor stress analysis



Rotor installed on the car

Bionic Wrench Manufacturing

DSGN 386 · Manufacturing Engineering & Design · 2025

We reverse-engineered a commercial Bionic Wrench, then built everything needed to manufacture it: a seven-version assembly fixture, a single-operator line, the SOP, FMEA and value stream map, and a plan to scale to one million wrenches a year.

7

fixture versions

3:58

pilot cycle time

1M / yr

planned volume

18 days

mapped lead time

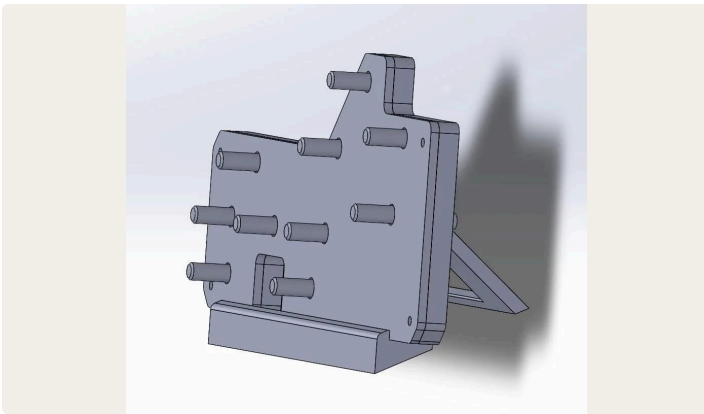
WHAT I DID

Led the fixture track design across all seven versions, from a flip-based locating fixture to the final angled, supported version.

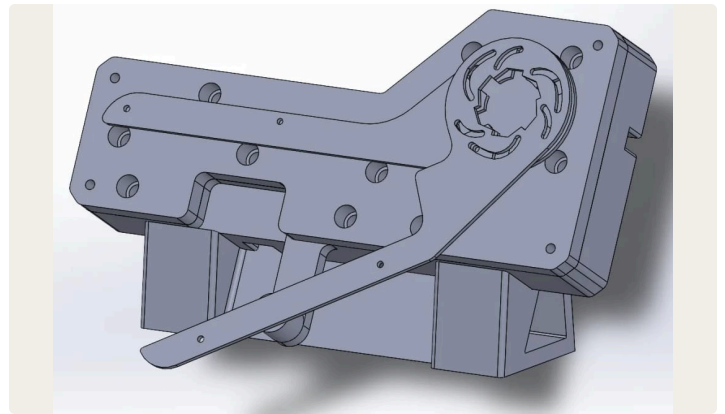
Drove the metrology and the CAD drawings; co-developed the SOP, FMEA and high-volume manufacturing plan.

Planned for seasonal demand: 6 s takt time and 25 parallel cells at peak, as few as 4 off-peak.

Pilot run: four good wrenches and two defective in a 30-minute timed trial (defects traced to out-of-spec jaws, not the process).



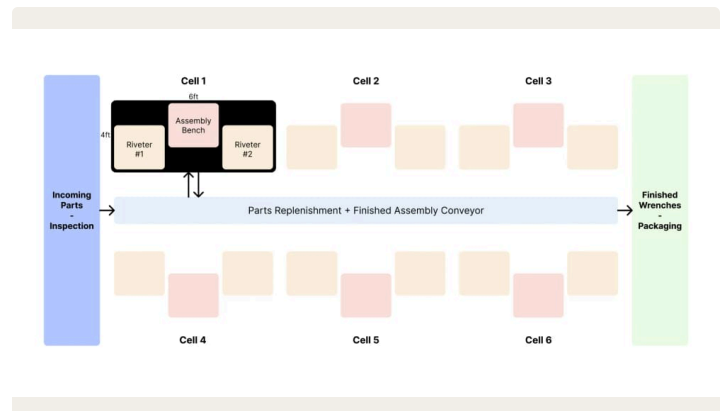
V2 fixture on its angled stand, with steel locating pins



V7, the final fixture



Single-operator line, equipment setup



Production line layout

Water Guard

Aquatic CT scanner waterproofing · 2025–2026

A removable waterproofing system that protects a ~\$1M portable CT scanner at a major Chicago aquarium while it images aquatic animals. Staff were spending 45+ minutes taping plastic sheeting before every scan; the scanner was never fully protected.

45 → <4 min

setup time per scan

~\$1M

scanner protected

1 liner

replaces plastic and tape

On-site

full-scale prototype

WHAT I DID

Led the gantry liner geometry (an oblique frustum shape) and the overlapping seam strategy.

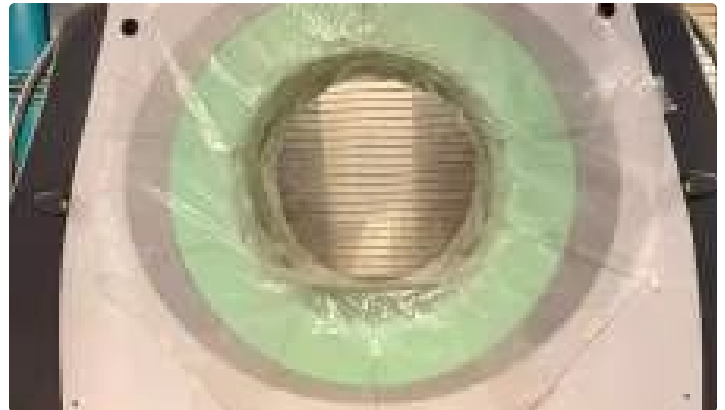
Ran saltwater degradation and tensile tests on heat-welded LDPE seams to choose the weld design.

Contributed to the FMEA and reliability analysis; synthesized the scanner manufacturer's engineering interviews into design constraints.

Assisted with fabrication of the full-scale prototype.



The scanner the system protects



Water Guard installed on the scanner



Full-scale slope test mockup



Weld seam leak test

Performance Bicycle Brake Caliper

Design, simulation and testing · 2025

A lightweight bicycle brake caliper designed to ISO 4210 safety requirements: translated needs into metrics, used FEA and topology optimization to cut mass, printed it in SLS Nylon-12, and tested it on a real bike.

35.5 g
final mass (−14%)

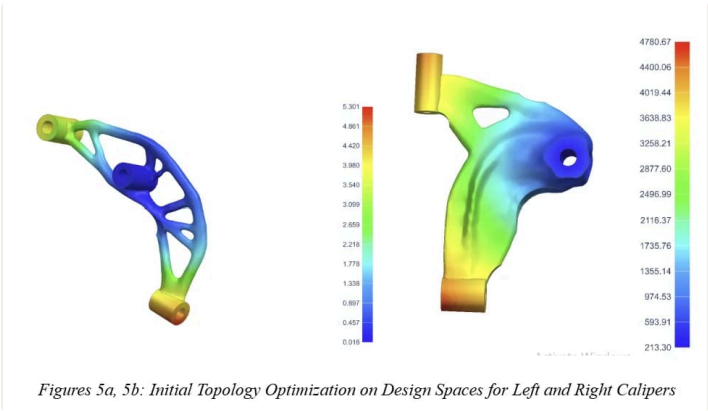
11.3 m
braking distance, iteration 1

\$8.26
cost per part

ISO 4210
safety standard

WHAT I DID

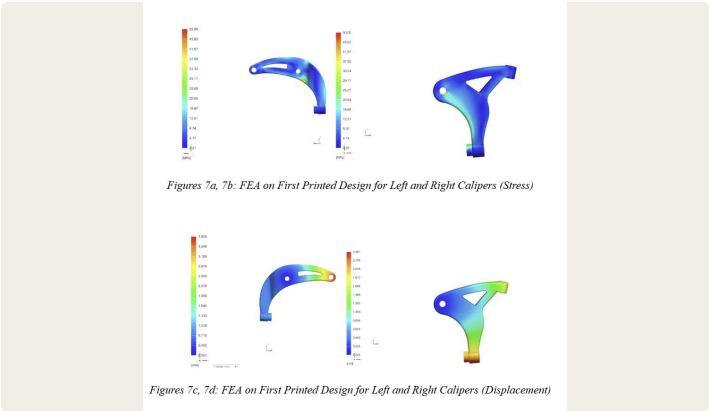
- Scheduling lead for the team.
- Led the stiffness-versus-mass trade-off evaluation and contributed to the FEA iteration decisions.
- Helped reduce the design space and validate the prototype on the bicycle.



Topology optimization



SLS Nylon-12 caliper



FEA validation of the first design

#	Need	Importance
1	The brake is lightweight.	4
2	The brake stops the bike within a safe distance.	5
3	The brake can be used with reasonable inputted force.	5
4	The brake is aerodynamic.	2
5	The brake is adjustable to an effective operating position.	4
6	The brake is aesthetically pleasing.	1
7	The brake is durable.	4
8	The brake is easily installed.	2
9	The brake is affordable.	3
10	The brake is safe in a crash.	4
11	The brake allows for easy pad replacement.	1
13	The brake resists environmental wear.	4
14	The brake can operate in a variety of weather conditions.	4
15	The brake does not fail with excessive input force.	5
16	The brake can be easily accessed for pad maintenance.	1
17	The brake should attach to standard bike mounting points.	5
18	The brake should allow proportional control of force.	4
19	The brake should be usable in a range of weather conditions.	3
20	The brake's parts operate with minimal friction.	4
21	The brake is usable with a variety of wheel rim widths.	3
22	The brake defaults to a position that does not exert force on the wheel.	4
23	The brake should be able to withstand the heat created during	4

Performance targets derived from the safety standard

Smart Sheet Smith

Advanced Intelligent Manufacturing Lab · NSF ERC-HAMMER · 2026

A five-agent AI system that turns a 2D sheet-metal drawing into a verified bending process plan, with no 3D CAD model and no labeled data. Presented as a poster at MSEC 2026 / NAMRC54; I'm a co-author.

5

cooperating AI agents

50

drawings in the SIMBA dataset

6

judge metrics

Hundreds

drawings generated by script

WHAT I DID

Built the evaluation data for the Tracer Agent: CAD parts, multi-view drawings and hand-derived bend math.

Hand-labeled ground truth, logged the Tracer's failure modes, and reviewed Engineering Agent output as the manufacturing engineer (K-factor, inner-radius bend deduction, V-die ranges).

Wrote a SolidWorks VBA pipeline that randomizes a part's dimensions and exports hundreds of valid drawings, cutting data creation from days to minutes.

Gathered and sorted the engineering references for the system's knowledge base.

Vision Agent



Tracer Agent



Geometric Judge



Engineering Agent



Performance Judge



The Smart Sheet Smith poster at MSEC 2026 / NAMRC54

ExtendIt

Lecture-hall desk extension · 2024

A permanent, hinged desk extension for cramped lecture-hall desks, developed from user interviews through mockups to a load-tested prototype.

+50%

usable desk surface

8 / 10

users preferred it

30+ lb

held, under 1/2" deflection

7

physical mockups tested

WHAT I DID

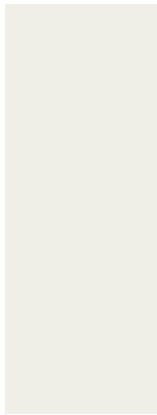
Ran 22 user interviews and led opportunity evaluation and the alternatives matrix.

Contributed to the hinge selection and mechanical design; built the load-testing rig and performed the structural tests.

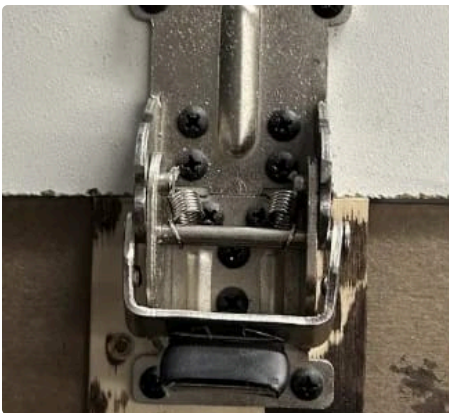
Helped develop the business model and cost structure.



Final prototype in a lecture hall



Load and deflection testing



Locking hinge under the desk

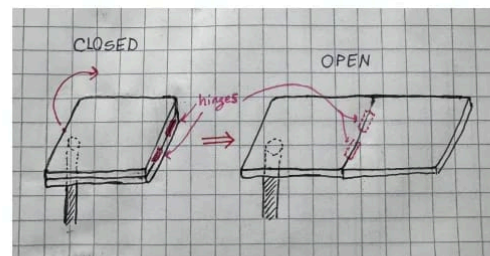


Figure 17- desk extension hinging sketch

Early hinge sketch

In the Shop

Machining, brazing, forming, casting, and repair

I learn by making. Most of my projects run through a shop: the mill and surface grinder, the torch, the waterjet, the foundry, and the bench.



Assembling a mechanism at the bench



Brazing copper tubing



Torch-brazing a joint



Hand-forming metal



Molten iron at a casting pour



Abrasive waterjet cutting



Working under a Tesla at a repair shop



Milling brake rotors

MORE

There's more online

Full write-ups, more projects, and a photo gallery

The site has a complete write-up for each project here, plus the work below, with more photos, drawings, and test data.

strasasher.github.io/Portfolio

ALSO ON THE SITE

Flexible Tactile Sensors

Resistive soft sensors and near-field electrospinning

P1 Motor Club Track Model

Large 3D-printed model of a racetrack property

SME Dorm Accessibility Project

Competition team lead, in concept development

Midwest EV Openpilot Retrofit

Enclosure design for a Tesla Model S retrofit

Drivetrain Efficiency Test Rig

Baja SAE CVT and gearbox loss measurement

StimSpin, Patchwork Plush, Iron Sand Casting

and more design and fabrication work

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